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Physical activity for the management of obesity in children up to the age of 9 years (Review)

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[Intervention Review]

Physical activity for the management of obesity in children up to the age of 9 years

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ABSTRACT

Rationale

Childhood obesity is a major global public health concern. Although physical activity is recognised as an effective non-pharmacological intervention, most existing evidence synthesis has primarily focused on the role of physical activity in preventing obesity, with limited attention given to its effects on managing children with obesity.

Objectives

To synthesise evidence on the benefits and harms of physical activity for the management of obesity in children up to 9 years of age.

Search methods

We searched CENTRAL, MEDLINE, Embase, and two trial registries from 2012 to 2 June 2023, with an update on 4 December 2025. We also used Google Scholar to identify additional studies. We did not impose language or publication status restrictions.

Eligibility criteria

We included randomised controlled trials (RCTs) that investigated the effects of physical activity interventions of any frequency, mode (e.g. aerobic, resistance), or intensity, with a minimum intervention duration of 12 weeks, in children up to 9 years of age with obesity (as defined by the trialists) at baseline. Eligible comparators were standard care, waiting-list, no physical activity, or any other active interventions, such as comparisons of different physical activity parameters (e.g. duration, frequency, intensity).

Outcomes

Critical outcomes: body mass index (BMI), BMI z-score, body weight, health-related quality of life, adiposity and fat distribution, glycaemia, and adverse events (minor and serious).

Important outcomes: physical well-being, mental well-being, physical activity levels, blood pressure, hyperinsulinemia, resistance to insulin, alterations in lipid metabolism, presence of obesity-related comorbidities or any non-communicable diseases, obesity-associated disability, lipid hormones, alterations in hunger or satiety, disability, mortality, prevalence of obesity in adulthood, and access to health services.

Risk of bias

Pairs of review authors independently assessed the risk of bias in included studies using the original version of the Cochrane tool (RoB 1).

Synthesis methods

We synthesised results using meta-analysis when appropriate. Given the clinical and statistical heterogeneity, we predominantly used a random-effects model alongside sensitivity analyses. When meta-analysis was not feasible, we employed synthesis without meta-analysis (SWiM) methods. We assessed the certainty of the evidence using the GRADE approach.

Included studies

We included four studies (five references, 517 children; female 46%; mean age ranged from 8.9 to 9.9) conducted in three countries (the USA, Brazil, and Iran). Three studies (75%) delivered the physical activity interventions in school settings. Three studies (75%) compared physical activity interventions with non-exercise controls, while one compared physical activity with a behaviour-changing intervention. The duration of physical activity interventions ranged from 12 to 32 weeks. We assessed three studies as having an overall high risk of bias.

Synthesis of results

Physical activity interventions compared to control

The evidence is very uncertain about the effect of physical activity interventions on BMI (mean difference (MD) −1.52 kg/m², 95% confidence interval (CI) −2.74 to −0.29; I^2 = 0%; 2 studies, 118 children; very low-certainty evidence), BMI z-scores (MD −0.10 z-score units, 95% CI −0.22 to 0.02; I^2 = 29%; 1 study, 222 children; very low-certainty evidence), body weight (MD −0.86 kg, 95% CI −3.17 to 1.46; I^2 = 0%; 2 studies, 118 children; very low-certainty evidence), health-related quality of life (MD −0.60 points, 95% CI −4.22 to 3.02; 1 study, 175 children; very low-certainty evidence), adiposity and fat distribution, assessed as body fat percentage (MD −1.23%, 95% CI −2.50 to 0.03; I^2 = 0%; 3 studies, 456 children; very low-certainty evidence), minor adverse events (risk ratio (RR) 3.58, 95% CI 1.95 to 6.55; 1 study, 222 children; very low-certainty evidence), and serious adverse events (RR 1.08, 95% CI 0.10 to 11.76; 1 study, 222 children; very low-certainty evidence). No studies assessed glycaemia.

Combined training (aquatic exercises) versus combined training (video game exercises)

The evidence is very uncertain about the effect of combined training in an aquatic setting compared to video-game-based combined training on BMI (MD −0.90 kg/m², 95% CI −3.21 to 1.41; 1 study, 39 children; very low-certainty evidence), BMI at 4-week follow-up (MD 0.03 kg/m², 95% CI −2.54 to 2.60; 1 study, 39 children; very low-certainty evidence), body weight (MD −1.30 kg, 95% CI −5.66 to 3.06; 1 study, 39 children; very low-certainty evidence), and body weight at 4-week follow-up (MD −0.70 kg, 95% CI −5.70 to 4.30; 1 study, 39 children; very low-certainty evidence). This study did not assess the other critical outcomes.

Low-dose versus high-dose combined training

The evidence is very uncertain about the effect of low-dose versus high-dose combined training on BMI z-scores (MD 0.12 z-score units, 95% CI 0.10 to 0.14; 1 study, 144 children; very low-certainty evidence), adiposity and fat distribution, assessed as body fat percentage (MD 1.07%, 95% CI −0.74 to 2.88; 1 study, 144 children; very low-certainty evidence), glycaemia, assessed as fasting glucose level (MD −0.50 mg/dL, 95% CI −2.77 to 1.77; 1 study, 144 children; very low-certainty evidence), minor adverse events (RR 0.91, 95% CI 0.64 to 1.30; 1 study, 144 children; very low-certainty evidence), and serious adverse events (RR 3.08, 95% CI 0.13 to 74.45; 1 study, 144 children; very low-certainty evidence). This study did not assess the other critical outcomes.

Authors' conclusions

The evidence from four randomised studies on the effects of physical activity interventions in children aged 0 to 9 with obesity is of very low certainty. Serious methodological limitations, clinical heterogeneity, small-study effects, and imprecise results constrained this evidence base. Important knowledge gaps remain because none of the included RCTs enroled children with disabilities. The RCTs provided little information on contextual factors. Future high-quality and better-reported RCTs will likely change our findings.

Funding

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Registration

Protocol available via https://doi.org/10.17605/OSF.IO/DSHUP

PLAIN LANGUAGE SUMMARY

What are the effects of physical activity for children with obesity?

Key messages

- Compared to no physical activity, any type of physical activity may help children with obesity to reduce their BMI slightly (the 'body mass index' is used to check if someone is a healthy weight) and may lead to considerably more minor unwanted events, but we are very uncertain about these results.
- We do not know if physical activity has any effect on the weight, body fat percentage, well-being, and blood sugar levels of children with obesity compared to no physical activity or another type or amount of physical activity. We also do not know if physical activity has any effect on the number of serious unwanted effects.
- We need larger, longer-lasting, and better-quality studies to be more certain about the effects of physical activity on children with obesity.

What is obesity?

Obesity is when a person has too much body fat. Worldwide, obesity amongst children (aged 0 to 9 years) has increased substantially since 1975. More than 300 million children and adolescents (aged 10 to 19 years) may be living with obesity by 2050.

Children with obesity are more likely to develop health problems such as heart disease, stroke, type 2 diabetes, some cancers, liver disease, asthma, joint and muscle problems, and infections later in life. Childhood obesity is also linked to higher rates of anxiety and depression.

Weight loss is the best way to reduce the health risks caused by obesity. In children, this is usually done through lifestyle changes such as doing more physical activity (e.g. running games or interactive video games that encourage movement (sometimes called 'exergaming')) and eating healthier foods.

What did we want to find out?

We wanted to find out if, compared to no physical activity, physical activity helps children with obesity:

- reduce their BMI (the 'body mass index' indicates if someone is a healthy weight; their weight is divided by their height multiplied by itself, expressed as kg/m^2) or BMI z-score (how someone's BMI compares to BMIs of others of the same age and sex);
- lose weight;
- improve their well-being;
- reduce body fat percentage or improve how body fat is distributed in the body; and
- lower their blood sugar levels.

We also wanted to find out if one type or amount of physical activity was better than another type or amount at producing these benefits, and if the physical activity programmes led to any unwanted effects.

What did we do?

We searched for studies that compared participating in physical activity to:

- no physical activity;
- another type or amount of physical activity.

The children needed to do the physical activities for at least 12 weeks. We compared and summarised the results of the studies and rated our confidence in the evidence, based on factors such as study methods and number of participants.

What did we find?

We found four studies including 517 children. The studies were small: the smallest included only 59 children and the largest, 222. Just under half of the children involved were girls (46%). Their average age was 9 years.

In three studies, the children did the physical activities for 12 or 13 weeks; the fourth study's physical activity programme lasted 32 weeks. The studies took place in three countries: one study each in Brazil and Iran, and two studies in the USA. In the American studies, 73% of the children were Black. The other studies did not describe the children's ethnicities.

- All four studies compared physical activity to no physical activity.
- One study also compared one type of physical activity (exergaming) to another type (swimming pool exercises).
- One study also compared doing 20 minutes versus 40 minutes of the same physical activities.

Main results

Physical activity versus no physical activity

Compared to no physical activity, children with obesity who participated in any physical activity programme may have:

- slightly reduced their BMI: the BMI of children in the active groups was on average 1.52 kg/m^2 lower (2 studies, 118 children);
- experienced more minor unwanted effects: children in the active groups may have been about 3.5 times more likely to experience minor unwanted effects (1 study, 222 children).

However, we are very uncertain about these results.

We do not know if physical activity makes any difference to children's:

- BMI z-score;
- weight;
- well-being;
- body fat percentage or how fat is distributed in the body;
- blood sugar levels;
- risk of experiencing serious unwanted events.

One type or amount of physical activity versus another type or amount

We are very uncertain about the effect of one type or amount of physical activity compared with another type or amount on any of the points we were interested in.

What are the limitations of the evidence?

We have limited confidence in the evidence because:

- there were few studies and they were small;
- the types of physical activities varied;
- the studies' reporting of their methods and findings was poor;
- three out of four studies did not report all their results.

How up to date is this evidence?

This evidence is current to December 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Physical activity (any form) compared to control (non-exercise or behaviour change) for children aged up to 9 years with obesity

Physical activity (any form) compared to control (non-exercise or behaviour change) for children aged up to 9 years with obesity

Population: children aged up to 9 years with obesity in all their diversity

Setting: schools (3 RCTs) and gym (1 RCT)

Intervention: physical activity (any form)

Comparison: control (non-exercise or behaviour change)

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with control (non-exercise or behaviour change)	Risk with physical activity (any form)				
BMI (kg/m^2) Assessed at end of intervention (3 months) Lower values are better	The mean BMI (kg/m^2) was **28 kg/m^2**	MD **1.52 kg/m2 lower** (2.74 lower to 0.29 lower)	-	118 (2 RCTs)	⊕◯◯◯ Very low[a,b]	Compared to control, physical activity may reduce BMI, but the evidence is very uncertain: relative change 5% lower (95% CI 10% lower to 1% lower).
BMI z-score Assessed at end of intervention (3 to 4 months) Lower values are better	The mean BMI z-score was **2.6 SD**	MD **0.1 SD lower** (0.22 lower to 0.02 higher)	-	222 (1 RCT)	⊕◯◯◯ Very low[c,d]	The evidence is very uncertain about the effect of physical activity on BMI z-score: relative change 5% lower (95% CI 11% lower to 1% higher).
Body weight (kg) Assessed at end of intervention (3 months) Lower values are better	The mean body weight (kg) was **40 kg**	MD **0.86 kg lower** (3.17 lower to 1.46 higher)	-	118 (2 RCTs)	⊕◯◯◯ Very low[c,e]	The evidence is very uncertain about the effect of physical activity on body weight: relative change 2% lower (95% CI 8% lower to 4% higher).

Health-related quality of life Assessed at end of intervention (8 months), with PedsQL child self-report: 23 items that yield total, physical summary, and psychosocial summary scores, each with a possible range of 0 to 100 Higher values are better		The mean health-related quality of life was **81 points**	MD **0.6 points lower** (4.22 lower to 3.02 higher)	-	175 (1 RCT)	⊕◯◯◯ Very low[f,g]	The evidence is very uncertain about the effect of physical activity on health-related quality of life: relative change 8% lower (95% CI 57% lower to 41% higher).
Adiposity and fat distribution Assessed as body fat percentage, at end of intervention (3 to 8 months) Lower values are better		The mean body fat percentage was **39%**	MD **1.23% lower** (2.5 lower to 0.03 higher)	-	456 (3 RCTs)	⊕◯◯◯ Very low[h,i]	The evidence is very uncertain about the effect of physical activity on body fat percentage: relative change 1% lower (95% CI 7% lower to 4% higher).
Glycaemia Assessed as fasting glucose levels		Outcome not reported					
Adverse events Assessed at end of intervention (3 to 4 months) Fewer events are better	**Minor adverse events**	128 per 1000	**459 per 1000** (250 to 840)	**RR 3.58** (1.95 to 6.55)	222 (1 RCT)	⊕◯◯◯ Very low[a,j]	Compared to control, physical activity may increase the risk of minor adverse events, but the evidence is very uncertain: relative change 257% more (95% CI 95% more to 555% more); NNTH 3 (95% CI 1 to 8).
	Serious adverse events	13 per 1000	**14 per 1000** (1 to 151)	**RR 1.08** (0.10 to 11.76)	222 (1 RCT)	⊕◯◯◯ Very low[a,k]	The evidence is very uncertain about the effect of physical activity (any form) on serious adverse events: relative change 8% more (95% CI 90% less to 1076% more).[l]

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

BMI: body mass index; **CI:** confidence interval; **MD:** mean difference; **NNTH:** number needed to treat for an additional harmful outcome; **PedsQL:** Pediatric Quality of Life Inventory; **RCT:** randomised controlled trial; **RR:** risk ratio; **SD:** standard deviation

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_c26b4b53-e406-4820-9364-ebc417aff534-1783163430314.

[a]Downgraded by two levels because of unclear to high risk of bias (selection, reporting, and other bias).
[b]Downgraded by one level due to imprecision, as the number of children was limited.
[c]Downgraded by one level because of unclear to high risk of bias (selection, reporting, and other bias).
[d]Downgraded by two levels due to imprecision: the 95% confidence interval crosses the minimal clinically important difference of 0.25 SD, and the number of children was limited. We assumed a minimal clinically important between-group difference of 0.25 SD on the BMI z-score.
[e]Downgraded by two levels due to imprecision: the 95% confidence interval includes potential benefit and harm, and the number of children was limited.
[f]Downgraded by one level because of unclear to high risk of bias (selection, detection, and reporting bias).
[g]Downgraded by two levels due to imprecision: the 95% confidence interval crosses the minimal clinically important difference of 4.36, and the number of children was limited. We assumed a minimal clinically important between-group difference of 4.36 raw units for the PedsQL child's self-report.
[h]Downgraded by two levels because of unclear to high risk of bias (selection, performance, detection, reporting, and other bias).
[i]Downgraded by one level due to imprecision: the 95% confidence interval includes potential benefit and harm, and the number of children was limited.
[j]Downgraded by one level due to imprecision, as there were few events reported.
[k]Downgraded by two levels due to imprecision: the 95% confidence interval includes potential benefit and harm, and there were few events reported.
[l]We did not calculate NNTH as the outcome did not show a between-group difference.

Summary of findings 2. Combined training (aquatic exercises) compared to combined training (video game exercises) for children up to the age of 9 years with obesity

Combined training (aquatic exercises) compared to combined training (video game exercises) for children up to the age of 9 years with obesity

Population: children up to the age of 9 years with obesity in all their diversity

Setting: school

Intervention: combined training (aquatic exercises)

Comparison: combined training (video game exercises)

Outcomes		Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
		Risk with combined training (video game exercises)	Risk with combined training (aquatic exercises)				
BMI (kg/m^2) Lower values are better	Assessed at end of intervention (3 months)	The mean BMI (kg/m^2) was **27.36 kg/m^2**	MD **0.9 kg/m^2 lower** (3.21 lower to 1.41 higher)	-	39 (1 RCT)	⊕◯◯◯ Very low[a,b]	The evidence is very uncertain about the effect of combined training (aquatic or video game exercises) on BMI: relative change 3% lower (95% CI 12% lower to 5% higher).

	Assessed at 4-week follow-up	The mean BMI (kg/m^2) was **27 kg/m^2**	MD **0.03 kg/m^2 higher** (2.54 lower to 2.6 higher)	-	39 (1 RCT)	⊕○○○ Very low[a,b]	The evidence is very uncertain about the effect of combined training (aquatic or video game exercises) on BMI at 4 weeks of follow-up: relative change 0.1% higher (95% CI 9% lower to 10% higher).
BMI z-score	Outcome not reported						
Body weight (kg) Lower values are better	Assessed at end of intervention (3 months)	The mean body weight (kg) was **36.8 kg**	MD **1.3 kg lower** (5.66 lower to 3.06 higher)	-	39 (1 RCT)	⊕○○○ Very low[a,b]	The evidence is very uncertain about the effect of combined training (aquatic or video game exercises) on body weight: relative change 3% lower (95% CI 15% lower to 8% higher).
	Assessed at 4-week follow-up	The mean body weight (kg) was **36.8 kg**	MD **0.7 kg lower** (5.7 lower to 4.3 higher)	-	39 (1 RCT)	⊕○○○ Very low[a,b]	The evidence is very uncertain about the effect of combined training (aquatic or video game exercises) on body weight at 4 weeks of follow-up: relative change 2% lower (95% CI 15% lower to 12% higher).
Health-related quality of life	Outcome not reported						
Adiposity and fat distribution	Outcome not reported						
Glycaemia	Outcome not reported						
Adverse events	Outcome not reported						

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

BMI: body mass index; **CI:** confidence interval; **MD:** mean difference

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

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[a]Downgraded by one level because of unclear to high risk of bias (selection, reporting, and other bias).
[b]Downgraded by two levels due to imprecision: the 95% confidence interval includes potential benefit and harm, and the number of children was limited.

Summary of findings 3. Combined training (low dose) compared to combined training (high dose) for children up to the age of 9 years with obesity

Combined training (low dose) compared to combined training (high dose) for children up to the age of 9 years with obesity

Population: children up to the age of 9 years with obesity in all their diversity

Setting: gym

Intervention: combined training (low dose)

Comparison: combined training (high dose)

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with combined training (high dose)	Risk with combined training (low dose)				
BMI (kg/m^2)	Outcome not reported					
BMI z-score Assessed at end of intervention (3 to 4 months) Lower values are better	The mean BMI z-score was **2.02 SD**	MD **0.12 SD higher** (0.1 higher to 0.14 higher)	-	144 (1 RCT)	⊕◯◯◯ Very low[a,b]	The evidence is very uncertain about the effect, but compared to combined training at low dose, combined training at high dose may reduce BMI z-score: relative change 6% higher (95% CI 5% higher to 7% higher).
Body weight (kg)	Outcome not reported					
Health-related quality of life	Outcome not reported					
Adiposity and fat distribution Assessed as body fat percentage, at end of intervention (3 to 4 months)	The mean body fat (%) was **39.11 %**	MD **1.07 % higher** (0.74 lower to 2.88 higher)	-	144 (1 RCT)	⊕◯◯◯ Very low[a,c]	The evidence is very uncertain about the effect of combined training at high or low dose on body fat percentage: relative change 3% higher (95% CI 2% lower to 7% higher).

Lower values are better							
Glycaemia Assessed as fasting glucose levels (mg/dL), at end of intervention (3 to 4 months) Lower values are better		The mean fasting glucose was **91.8 mg/dL**	MD **0.5 mg/dL lower** (2.77 lower to 1.77 higher)	-	144 (1 RCT)	⊕◯◯◯ Very low[a,c]	The evidence is very uncertain about the effect of combined training at high or low dose on fasting glucose: relative change 0.5% lower (95% CI 3% lower to 2% higher).
Adverse events Assessed at end of intervention (3 to 4 months) Fewer events are better	**Minor adverse events**	479 per 1000	**436 per 1000** (307 to 623)	**RR 0.91** (0.64 to 1.30)	144 (1 RCT)	⊕◯◯◯ Very low[a,d]	The evidence is very uncertain about the effect of combined training at high or low dose on minor adverse events: relative change 9% less (95% CI 36% less to 30% more)[e].
	Serious adverse events	0 per 1000	**-- per 1000** (0 to 0)	**RR 3.08** (0.13 to 74.45)	144 (1 RCT)	⊕◯◯◯ Very low[a,d]	The evidence is very uncertain about the effect of combined training at high or low dose on serious adverse events: relative change 208% more (95% CI 87% less to 7.345% more)[e].

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

BMI: body mass index; **CI:** confidence interval; **MD:** mean difference; NNTH: number needed to treat for an additional harmful outcome; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

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[a]Downgraded by one level because of unclear to high risk of bias (selection, reporting, and other bias).
[b]Downgraded by two levels due to imprecision: the 95% confidence interval crosses the minimal clinically important difference of 0.25 SD, and the number of children was limited. We assumed a minimal clinically important between-group difference of 0.25 SD on the BMI z-score.
[c]Downgraded by two levels due to imprecision: the 95% confidence interval includes potential benefit and harm, and the number of children was limited.
[d]Downgraded by two levels due to imprecision: the 95% confidence interval includes potential benefit and harm, and there were few events reported.
[e]We did not calculate NNTH as the outcome did not show a between-group difference.

BACKGROUND

Childhood obesity is a major global public health concern [1]. Its prevalence has increased substantially over recent decades. Since 1975, the number of girls with obesity has increased from approximately 5 million to 50 million (95% credible interval 24 to 89 million), and among boys from 6 million to 74 million (95% credible interval 39 to 125 million) in 2016 [2]. The overall prevalence of childhood obesity increased from 0.7% to 7.8% over the same period [2]. In 2020, the World Health Organization (WHO) estimated that 39 million children under five years of age were diagnosed with overweight or obesity, with nearly 31.5 million living in low- and middle-income countries [1, 3].

Description of the condition

Obesity is defined as excess accumulation of body fat that poses a risk to health. Obesity results from complex interactions among socio-cultural, environmental, behavioural, genetic, and economic factors that contribute to positive energy imbalances [4]. Socioeconomic status is strongly associated with obesity, although the direction and magnitude of this relationship vary by setting and stage of economic development [5, 6]. In high-income countries, obesity is more prevalent among children from lower socioeconomic groups, whereas in some low- and middle-income countries the opposite pattern is observed, with higher prevalence among relatively wealthy families [5, 6]. Childhood obesity, particularly moderate and severe obesity, is also associated with wider socioeconomic consequences, including increased income inequality and intergenerational persistence [5]. Overall, the burden of obesity is not evenly distributed across populations, with the greatest prevalence often observed among socially and economically disadvantaged groups in high-income settings [5, 6]. Research has explored multiple interacting pathways linking socioeconomic position and obesity, but these relationships remain complex and context-dependent [7, 8].

Childhood obesity can affect both physical and mental well-being [9, 10]. It increases the risk of cardiovascular disease (such as stroke, aneurysm, and heart attack), diabetes, and some cancers (including uterine, oesophageal, liver, kidney, pancreatic, and colorectal cancers). It is also associated with conditions such as liver disease, asthma, musculoskeletal disorders, pain, and increased susceptibility to infections [9]. Childhood obesity also brings a higher risk of mental health conditions, including anxiety and depression, with reported increases in risk of 43% in girls and 33% in boys [10]. Cohort and systematic review evidence suggests that children with obesity are at increased risk of premature morbidity and early death in young adulthood compared with peers without obesity [11, 12].

BMI cut-offs are used to categorise weight status as underweight, healthy weight, overweight, or obesity [13, 14]. Some countries have developed national BMI-for-age reference standards for children, while others use international standards such as those developed by the WHO [15, 16 17, 18], as it has been shown that BMI z-score is the most useful measure in children. However, the different methods used to determine weight status or categories amongst children can make it difficult to compare the outcomes presented in the literature.

Description of the intervention and how it might work

Physical activity is defined as any bodily movement produced by skeletal muscles that results in energy expenditure (e.g. walking to school, helping with housework, children's games) [19]. Exercise training is a subtype of physical activity in which bodily movement is planned, structured, and repetitive, with the objective of improving or maintaining muscular and cardiorespiratory fitness and health [19, 20]. Exercise training can be further broken down into three types: aerobic, resistance, and combined (aerobic plus resistance) exercise training. Exercise training intervention characteristics may be described using the frequency, intensity, type, time, volume, and progression (FITT-VP) framework, which is widely used in exercise prescription and reflected in physical activity guidance from organisations such as the American College of Sports Medicine and the WHO [20, 21]. Exercise training interventions as well as broader physical activity interventions (e.g. recreational programmes) were eligible for inclusion in this review [22, 23].

Childhood obesity can be managed through everyday behavioural changes involving energy intake and energy expenditure [1, 24]. Exercise training prescription or dosage should aim to obtain a large negative energy balance (i.e. where the amount of calories expended exceeds the amount of energy consumed), long-term adherence, better health outcomes, and improved well-being. Epidemiological studies have shown that physical activity and exercise training have beneficial effects on obesity [25, 26].

The effects of physical activity and exercise training on weight status in young children are inconsistent across studies. For example, in a 2019 systematic review, Kelley and colleagues found that exercise training interventions were effective in preventing and managing obesity, decreasing BMI and body fat percentage in children and adolescents aged 8 to 17 with overweight and obesity [23]. However, in a 2019 Cochrane review, Brown and colleagues found evidence of no effects of physical activity on BMI z-score and BMI in young children (aged 0 to 5 years) [7]. Evidence from a 2019 umbrella review showed that exercise training (aerobic exercise, resistance training, and combined training) was effective in decreasing body weight, BMI, BMI z-score, central obesity, fat mass, visceral fat, and subcutaneous fat in children and adolescents with overweight and obesity aged 7 to 17 years of age [27]. Similarly, high-intensity interval training may reduce body weight compared to moderate-intensity continuous training in children and adolescents aged 6 to 18 years with overweight and obesity, potentially through modifications in energy balance [28]. The 2020 WHO guidelines found that physical activity may improve some measures of adiposity – particularly the sum of four skinfold thicknesses – but had little to no effect on waist circumference, hip circumference, or body fat percentage in children aged 7 to 12 years; however, the certainty in the evidence is very low [29]. According to the 2018 US Physical Activity Guidelines Advisory Committee, strong evidence demonstrates that higher levels of physical activity are associated with smaller increases in weight and adiposity in children aged 3 to 17 years [30].

Beyond weight-related outcomes, research has shown that physical activity and exercise training in children with obesity may also improve quality of life, academic achievement, cardiorespiratory fitness, and cognitive performance [31, 32, 33].

Why it is important to do this review

Despite substantial global efforts to prevent childhood overweight and obesity, millions of children worldwide are already living with obesity and require effective treatment strategies [1, 2, 3]. Much of the existing literature and policy focus has centred on prevention approaches or interventions targeting older children and adolescents, while comparatively less attention has been given to the management of obesity in younger children.

Physical activity is widely promoted as a key component of childhood obesity management, and a range of initiatives – at family, school, and community levels – has been implemented internationally to increase opportunities for physical activity and reduce obesogenic environments [1, 9]. However, uncertainty remains regarding the effectiveness of physical activity or exercise training interventions specifically for children aged 0 to 9 years who are already living with obesity. Younger children may differ from older children and adolescents in important developmental, behavioural, family, and environmental respects that could influence both the feasibility and effectiveness of interventions. A focused synthesis of the evidence in this age group is therefore important to inform clinical practice, public health policy, and future intervention development.

OBJECTIVES

To synthesise evidence on the benefits and harms of physical activity for the management of obesity in children up to 9 years of age.

METHODS

We attempted to address the following research question: in infants and children with obesity up to 9 years of age (population), what is the effect of physical activity (intervention) compared with no intervention or other interventions (comparisons), for the management of obesity to improve health, functioning, and to reduce obesity-associated disability (outcome)?

We followed guidance in the *Cochrane Handbook for Systematic Reviews of Interventions* and the Methodological Expectations for Cochrane Intervention Reviews (MECIR) manual when conducting the review [34, 35], and PRISMA 2020 for the reporting [36].

Differences between protocol and review

We revised our original eligibility criteria to include randomised controlled trials (RCTs) with a minimum intervention duration of 12 weeks, regardless of the post-intervention follow-up period.

Criteria for considering studies for this review

Types of studies

We included RCTs, including individually-randomised and cluster-randomised trials (randomising six or more groups of participants, i.e. at least three clusters per arm). We excluded cross-over trials. Eligible studies had to address physical activity interventions with a minimum duration of 12 weeks and be published in the year 2012 or later.

According to current National Institute for Health and Care Excellence (NICE) recommendations, multicomponent interventions, such as physical activity for obesity management, should be implemented for a minimum duration of three months (12 weeks) to allow sufficient time for behavioural change and measurable outcomes [37].

Types of participants

We included children aged 0 to 9 years with obesity as defined by the trialists (e.g. by BMI z-score or BMI percentiles) [38, 39, 40]. If studies involved infants and children within a family group receiving the intervention, and outcome data were reported separately for the infants and children, we included these studies.

In line with WHO principles, we planned to include children with obesity in all their diversity. "In all their diversity" entails recognising, accepting, and drawing strength from individual differences, including diversity in gender, disability, body size, nationality or geographic background, ethnicity, religion, family composition, and socioeconomic status.

Types of interventions

We included physical activity interventions with a minimum 12 weeks' duration, of any frequency, mode (e.g. aerobic exercise, resistance training, combined training), or intensity, for the management of obesity in children aged up to and including 9 years [29]. Digital interventions aiming to instruct and guide physical activity performance were eligible only if they were not limited to the promotion of active behaviour. We excluded RCTs of mixed interventions (e.g. combined diet/nutritional education and physical activity), as other reviews have addressed these approaches [8, 41].

Eligible comparators included standard care (e.g. pharmacological interventions, behaviour-changing interventions), waiting-list control, non-exercise interventions, or any other active interventions, such as different physical activity parameters (e.g. duration, frequency, patterns, types, or intensities [42, 43]).

Outcome measures

Following discussion with the Cochrane editorial team, and in accordance with MECIR guidance [34], we selected seven critical outcomes for inclusion in the summary of findings tables, and classified the remaining 15 outcomes as important outcomes (see below). We rated the certainty of the evidence for all 21 outcomes using the GRADE framework to facilitate their use in the guideline development context.

Given the broad scope of the review, we imposed no restrictions on either the eligible outcome measurements or the timing of measurement. We planned to use validated scales where available; if these were unavailable, we accepted outcome measures as defined and assessed by the trialists. We excluded studies that did not assess our outcome measures of interest. Our primary time point for outcome assessment was at the end of the intervention. We reported results at post-intervention follow-up time points when data were available.

Critical outcomes

- BMI (kg/m^2)
- BMI z-score
- Body weight (kg)
- Health-related quality of life

- Adiposity and fat distribution (measured, for example, by dual-energy X-ray absorptiometry (DXA), magnetic resonance imaging (MRI), bioelectrical impedance analysis, skinfold assessment, or other methods; including outcomes such as body fat percentage, fat mass, visceral adiposity, waist circumference, waist-to-hip ratio, sum of skinfold thicknesses, or other measures defined by trialists)
- Glycaemia (assessed using fasting blood glucose or other measures defined by trialists)
- Adverse events: minor and serious adverse events (as defined by trialists)

Important outcomes

- Physical well-being (e.g. functioning and mobility, self-care, getting along, life activities and participation, or as defined by trialists)
- Mental well-being (e.g. anxiety, stress, depression, isolation, self-esteem, body image, suicidal ideation, eating disorders, or as defined by trialists)
- Physical activity levels (as defined by trialists)
- Blood pressure
- Hyperinsulinemia (1.7 mmol/L; 30 mg/dL or higher when 1 mg of glucagon is administered intramuscularly/intravenously, or as defined by trialists)
- Resistance to insulin (as defined by trialists)
- Alterations in lipid metabolism or in adipogenesis and fat accumulation
- Presence of obesity-related comorbidities or any non-communicable diseases
- Obesity-associated disability
- Lipid hormones (e.g. leptin or adiponectin)
- Alterations in hunger or satiety (as defined by trialists)
- Disability in any of the functionality domains, according to the International Classification of Functioning, Disability and Health [44]
- Mortality
- Prevalence of obesity in adulthood
- Access to health services

Search methods for identification of studies

Electronic searches

We limited our literature searches to studies published from 2012 onwards because earlier studies had already been comprehensively screened in the 2017 Cochrane review by Mead and colleagues on the management of obesity in children [45]. We therefore considered restricting searches to studies published from 2012 onwards to be a reasonable and methodologically appropriate approach. None of the studies included in the Mead review met our inclusion criteria [45], as the two physical activity-focused studies within the eligible age range specifically targeted children with overweight rather than obesity.

An experienced information specialist (CMEL) developed the search strategy. Following a pilot search, we applied an adapted search strategy across all databases listed below from 2012 to 2 June 2023, with an updated search completed on 4 December 2025 (Supplementary material 1).

- Cochrane Central Register of Controlled Trials (CENTRAL; Issue 12 of 12, December 2025) in the Cochrane Library (from 2012 to 04 December 2025)
- Ovid MEDLINE ALL (2012 to 04 December 2025)
- Ovid Embase (2012 to 04 December 2025)
- ClinicalTrials.gov (www.clinicaltrials.gov; from 2012 to 04 December 2025)
- World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) (www.who.int/trialsearch; from 2012 to 04 December 2025)

Searching other resources

An external librarian peer-reviewed the search strategy. We applied no restrictions on language or place of publication. Further details of the search strategy are also available in our protocol (https://osf.io/anqb8). Additionally, we searched Google Scholar to identify grey literature (e.g. reports, dissertations, theses, and conference abstracts).

We also searched for post-publication amendments to eligible studies, including corrections, expressions of concern, and retractions, and we checked and incorporated any modifications, errors, or retractions up to the date of review submission.

Data collection and analysis

Selection of studies

We deduplicated the search results in EndNote and screened records in Covidence [46, 47]. Two review authors (from amongst AFLB, LEIG, NCG, CMEL, JB) independently piloted the screening forms. Working independently, two review authors screened titles, abstracts, and relevant full-text articles against the eligibility criteria. A third review author adjudicated disagreements. For studies reported in several documents, we identified and collated all related documents to ensure participants were only counted once.

Data extraction and management

Two review authors (from amongst AFLB, LEIG, NCG, CMEL, JB) independently extracted data using pre-standardised data collection forms. A third review author adjudicated disagreements.

Risk of bias assessment in included studies

Pairs of review authors (from amongst AFLB, LEIG, NCG, CMEL, JB) independently assessed risk of bias using the original version of the Cochrane tool for assessing risk of bias (RoB 1) [48]. We included the study funder in the 'Other bias' category of the RoB 1 tool. We collected and recorded information about authors' financial and non-financial conflicts of interest reported in the studies.

Blinding of physical activity-related interventions is challenging and, in most cases, not feasible. We assessed blinding separately for different outcomes where appropriate; for example, the risk of bias due to lack of blinding may differ between adverse events and patient-reported outcomes. We accounted for this in both the risk of bias and GRADE assessments by considering the likely impact of lack of blinding on the certainty of the evidence (e.g. subjective outcomes are likely to have a higher risk than objective outcomes).

Measures of treatment effect

For continuous outcomes, we calculated the absolute percent difference as the between-group difference in change from baseline (intervention minus control) expressed in the original measurement units. We derived the relative percent change from baseline by dividing the absolute effect by the pooled baseline mean of the intervention and control groups and expressing this as a percentage. We also calculated the relative difference in change from baseline by dividing the absolute effect by the baseline mean of the control group and expressing the result as a percentage.

For dichotomous outcomes, we expressed the relative effect as (risk ratio − 1) and reported this as a percentage change. We calculated the number needed to treat for an additional beneficial/harmful outcome (NNTB/H), as appropriate.

Unit of analysis issues

Although cluster-randomised trials were eligible for inclusion, we identified no cluster-RCTs meeting the review criteria. If we include studies with this design in future review updates, we will assess whether clustering was accounted for in the analyses. Where necessary, we will adjust for clustering using accepted methods and an appropriate intra-cluster correlation coefficient (ICC), obtained from study authors or similar studies where available [49, 50, 51].

Although many randomised trials have only two parallel arms (i.e. groups), some have three or four parallel arms; thus, a single randomised trial can yield several relevant comparisons. This review examined any relevant comparison that allowed evaluation of the effects of physical activity interventions on children aged 0 to 9 years. For example, a three-arm trial comparing physical activity (low and high intensity) versus no physical activity could appear in two separate analyses: low intensity versus no physical activity and high intensity versus no physical activity. If a control group was used as a comparator twice in the same analysis, we halved the sample size of the control group.

Dealing with missing data

When numerical data were missing, we contacted study authors to request the additional data required for analysis (Supplementary material 7).

If numerical data were available only in graphic form, we used Engauge version 5.1 to extrapolate means and standard deviations by digitising data points on the graphs [52].

When post-test standard deviations were unavailable, we used the standard deviations of the pre-test scores as estimates. If variance was expressed using statistics other than standard deviation (e.g. standard error, confidence interval, P value), we computed standard deviations according to the methods recommended in Chapter 6 of the *Cochrane Handbook for Systematic Reviews of Interventions* [53]. When missing standard deviations could not be derived via the methods described above, we imputed them from other studies in the meta-analysis.

Reporting bias assessment

To assess selective reporting within studies, we compared the outcomes prespecified in published protocols or trial registry records with those reported in the published study reports, including the number and order of outcomes where relevant. We searched the WHO International Clinical Trials Registry Platform and ClinicalTrials.gov to identify registry records corresponding to included studies. We documented trial registration numbers and the availability of published protocols in the risk of bias tables (see Supplementary material 2).

We assessed potential reporting bias between studies by constructing funnel plots when a sufficient number of studies (i.e. more than 10) were available for inclusion in a meta-analysis [54].

Synthesis methods

When two or more studies reported the same outcome, and we considered interventions to be sufficiently homogeneous, we pooled the data (meta-analysis) using Review Manager [55]. Before pooling data, we ensured consistency in the direction of effect across studies. Where necessary, we arithmetically reversed scale scores so that higher values consistently reflected the same direction of benefit. We standardised measurement units across studies to enable calculation of mean differences (e.g. 10-cm scales were expressed in mm to match other 100-mm scales). We presented results grouped by a common comparator; for example, exercise training versus control or high-intensity exercise training versus low-intensity exercise training. We included studies in the meta-analyses regardless of the risk of bias rating.

We planned to combine treatment effect estimates using inverse-variance weighting, utilising a random-effects model. We selected a random-effects model because we expected there to be clinical diversity in populations, interventions, outcome measures, study design, and settings [56, 57]. We estimated between-study variance using the restricted maximum likelihood (REML) method and calculated a confidence interval for the heterogeneity estimate [58]. We used the Hartung-Knapp-Sidik-Jonkman method to calculate confidence intervals for the pooled effect estimate when at least three studies were available and the heterogeneity estimate exceeded zero. In other scenarios (i.e. when only two studies were available or the heterogeneity estimate equalled zero), we used the Wald-type method. We calculated prediction intervals to provide a predicted range for the true treatment effect in an individual study [58]. We used a fixed-effect model for all analyses unless we considered heterogeneity to be evident, in which case we used a random-effects model and sensitivity analysis.

We followed the Synthesis Without Meta-analysis (SWiM) guideline to report effect data when we considered a meta-analysis to be inappropriate (e.g. unexplained high heterogeneity or effect estimates were incompletely reported) [59]. The SWiM guideline is a nine-item checklist to promote transparent reporting for reviews of interventions that use alternative synthesis methods.

Investigation of heterogeneity and subgroup analysis

We recognise that clinical and methodological heterogeneity may influence the interpretation of pooled results [60]. We first examined forest plots visually to assess the extent and possible sources of heterogeneity across studies. We then assessed statistical heterogeneity using the Chi^2 test (threshold $P < 0.10$), and quantified its magnitude using the following overlapping I^2 ranges described in the *Cochrane Handbook for Systematic Reviews of Interventions* [60]:

- 0% to 40%: might not be important;
- 30% to 60%: may represent moderate heterogeneity;

- 50% to 90%: may represent substantial heterogeneity;
- 75% to 100%: considerable heterogeneity.

As noted in the *Cochrane Handbook*, the importance of an observed value of I^2 depends on the magnitude and direction of effects, and the strength of evidence for heterogeneity (e.g. the P value from the Chi^2 test or a confidence interval for I^2). Uncertainty around I^2 is often substantial when the number of included studies is small.

Where sufficient data were available, we planned subgroup analyses to explore potential sources of heterogeneity according to participant characteristics (e.g. gender/sex, age, obesity severity, comorbidities, ethnicity, geographical region, nutritional status, and household income) and intervention characteristics (e.g. intervention duration, type and intensity of physical activity, and intervention setting). Consistent with guidance in the *Cochrane Handbook,* we planned to interpret subgroup findings cautiously, given the potential for multiple comparisons and spurious associations [60].

There were too few included studies to conduct these planned subgroup analyses.

Equity-related assessment

We designed this review to include infants and children in all their diversity and to consider broader social determinants of health. To examine potential differences in intervention effects across population groups, we explored whether the effects of physical activity interventions varied according to PROGRESS factors. The PROGRESS framework facilitates an explicit consideration of health equity-related factors when conducting, reporting, and adapting research for evidence-informed decision-making [61]. The framework includes **p**lace of residence, **r**ace/ethnicity/culture/language, **o**ccupation, **g**ender/sex, **r**eligion, **e**ducation, **s**ocioeconomic status, and **s**ocial capital.

We adapted our data extraction and evidence synthesis methods to identify any evidence regarding the mediating role of PROGRESS factors on the effects of physical activity interventions. See Supplementary material 8.

Sensitivity analysis

We planned to test the effects of risk of bias on the pooled effect estimates by removing studies that were at high or unclear risk of bias for the domains of blinding and incomplete outcome data. We planned to conduct sensitivity analyses of the ICCs using: (1) no adjustment; (2) adjustment for clustering assuming an ICC of 0.02; and (3) adjustment for clustering assuming an ICC of 0.04. However, we did not conduct formal sensitivity analyses to assess the impact of overall risk of bias, as too few trials were included for each outcome.

Certainty of the evidence assessment

We followed the GRADE approach to assess the certainty of the evidence and to prepare summary of findings (SoF) tables and evidence profiles for all prioritised outcomes [62]. In the SoF tables and evidence profiles, we presented the magnitude of intervention effects alongside the corresponding certainty of the evidence [63].

We adopted a partially contextualised approach to rate evidence certainty. Specifically, for each point estimate (or range of estimates), we assessed whether the true effect was likely to fall within predefined thresholds representing trivial, small, moderate, or large effects. We based these thresholds on the literature and refined them through discussions with clinical experts [63].

We interpreted pooled effect estimates as absolute and relative percentage changes for continuous outcomes and as relative risks with corresponding NNTB/NNTH values for dichotomous outcomes (see Measures of treatment effect).

The SoF tables present the seven prespecified critical outcomes (see Critical outcomes), corresponding effect estimates and risks, numbers of participants and studies, certainty ratings for each outcome, and explanatory footnotes and comments to aid readers' understanding.

We developed three SoF tables:

- physical activity versus no physical activity (Summary of findings 1)
- combined training (aquatic exercises) versus combined training (video game exercises) (Summary of findings 2);
- combined training (low dose) versus combined training (high dose) (Summary of findings 3).

We present the important outcomes in evidence profile tables (see Supplementary material 9, Supplementary material 13; Supplementary material 14; Supplementary material 15). This approach allowed us to report our certainty in the evidence for all outcomes.

We established a checklist to aid consistency and reproducibility of GRADE assessments for all comparisons (see Supplementary material 10; Supplementary material 11; Supplementary material 12) [64].

Consumer involvement

Members of the Guideline Development Group collaborated with the lead methodological team at the World Health Organization in prioritising the critical and important outcomes for this systematic review. They did not participate in the synthesis of findings from the included studies. The guideline panel received the SoF tables and discussed them with the review team. No other stakeholders or consumers were involved in the conduct of this systematic review.

RESULTS

Description of studies

Results of the search

This Cochrane review accompanies a separate 'sister' review titled 'Physical activity for the management of obesity in adolescents aged 10 to 19 years' [65]. We conducted a single overarching search-and-selection process to identify eligible studies for both reviews.

Our systematic search was conducted in two phases: an initial search on 2 June 2023 and a subsequent update on 4 December 2025. Our combined searches identified a total of 13,834 records, including 11,320 from bibliographic databases and 2514 from trial registries. After removing 3958 duplicates, we screened 9876 records and excluded 9587. This left 289 studies for full-text assessment. We excluded 255 studies, leaving 34 studies that met our inclusion criteria. We included four studies (five references) in the quantitative synthesis (Davis 2012 [66]; Irandoust 2020 [67]; Nobre 2017 [68]; Williams 2019 [69, 70]), and listed 29 studies (29 references) as 'awaiting classification' (seven conference abstracts, 12 trial registry records, five studies without full texts, and five studies written in languages other than those known to our team (three Persian and two Chinese studies)).

Six new trial registry records identified in the updated search met our inclusion criteria, with participant ages overlapping the eligible age ranges for both this review and its sister (i.e. 6 to 16 years old). We categorised these records as 'awaiting classification'. From the original search, we contacted the study investigators of 23 'awaiting classification' studies and have received two responses to date. Further details of the studies awaiting classification are provided in Supplementary material 4.

The study selection process is shown in Figure 1, and the characteristics of included studies are presented in Supplementary material 2.

Figure 1. PRISMA study flow diagram

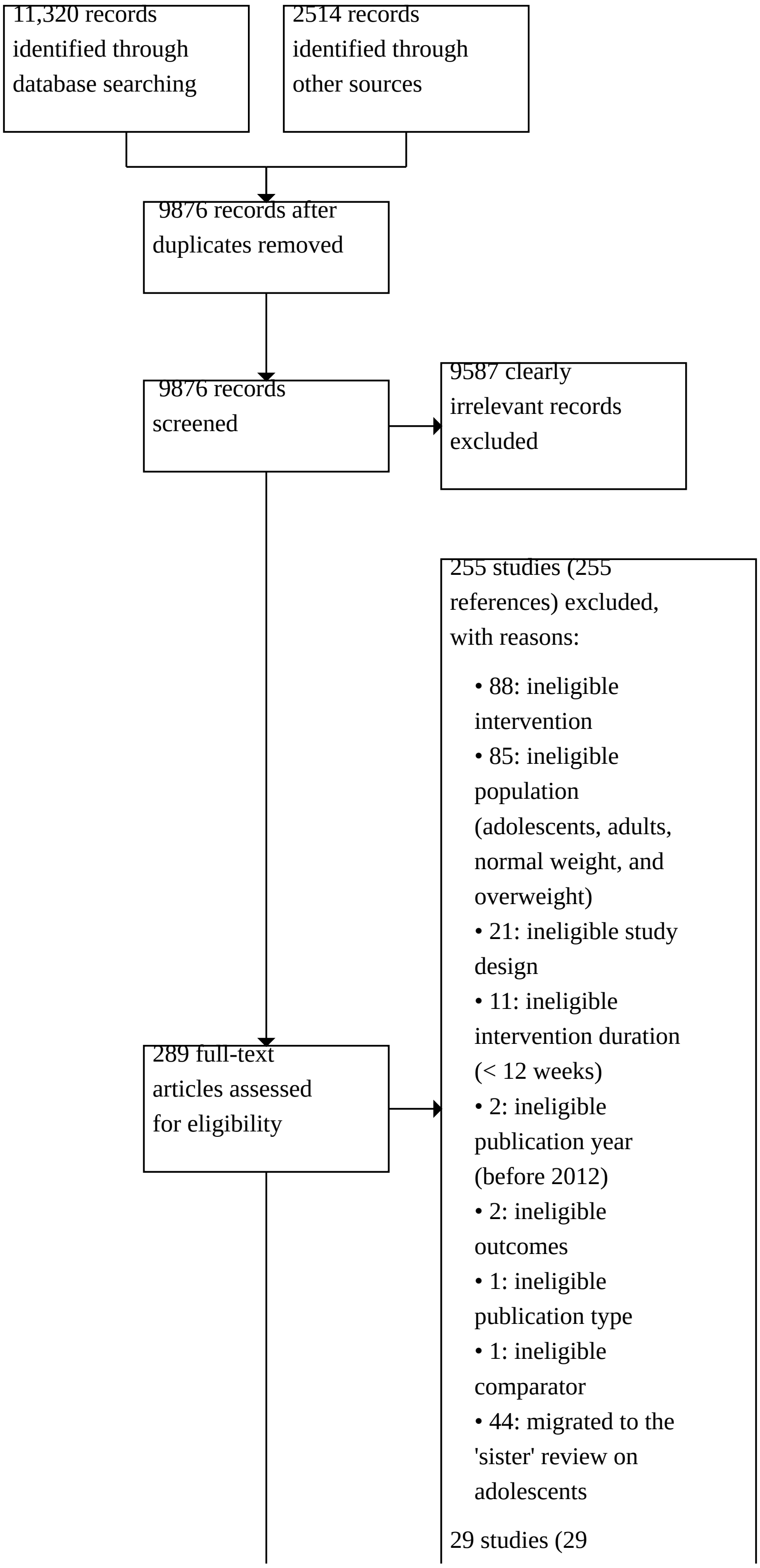
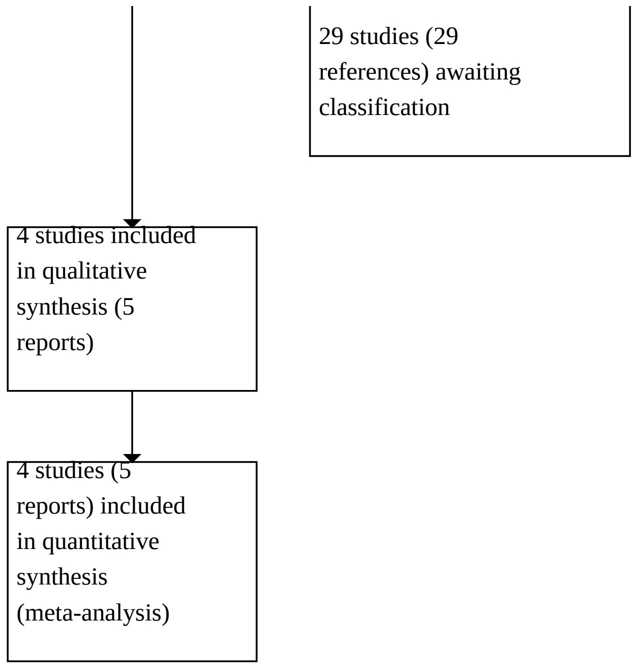


Figure 1. (Continued)



Included studies

We obtained most data presented in the review from published literature, including supplementary published data, information contained in clinical trial registers, and companion articles. For Williams 2019, we obtained additional information via correspondence with the study authors.

We included four parallel-group RCTs conducted in three countries: the USA (two studies: Davis 2012; Williams 2019), Brazil (one study: Nobre 2017), and Iran (one study: Irandoust 2020). They were published between 2012 and 2020 (median publication year 2016). All four studies reported their funding sources, and two provided trial identifiers (Davis 2012; Williams 2019). See Table 1 and Supplementary material 2.

Types of participants

The studies included a total of 517 children, with a mean age ranging from 8.9 years in Irandoust 2020 to 9.9 years in Nobre 2017. The number of children in the studies ranged from 59 (Nobre 2017) to 222 (Davis 2012). All studies reported the children's gender, with females accounting for 46% of the participants (Davis 2012; Irandoust 2020; Nobre 2017; Williams 2019). Two studies reported ethnicity (on average, 73% were Black children in both RCTs) (Davis 2012; Williams 2019).

The included studies utilised varying BMI thresholds for participant eligibility. The criteria used to diagnose obesity were a BMI above the 95th percentile in Irandoust 2020, and a BMI at or above the 85th percentile in Davis 2012 and Williams 2019. Nobre 2017 specified both BMI percentile diagnostic criteria (i.e. BMI ≥ 85th to 95th percentile and > 95th percentile). Regardless of the inclusion criteria, baseline BMI values for all studies indicate all participants were children with obesity (> 95th percentile) at the start of the trials (Davis 2012; Irandoust 2020; Nobre 2017; Williams 2019).

Only one study reported comorbidities, including hypertension, prediabetes, high cholesterol, and high triglycerides (Williams 2019).

Types of interventions and comparisons

Three studies (75%) used a 'no exercise training intervention' as a control group (Davis 2012; Irandoust 2020; Nobre 2017), and the fourth study compared physical activity to sedentary activities as a form of behaviour-changing intervention (Williams 2019).

Two studies had more than one active arm (i.e. three-arm trials) (Davis 2012; Irandoust 2020). One study (25%) had aerobic exercise as an intervention (Williams 2019), and three studies (75%) had combined training (Nobre 2017; Davis 2012; Irandoust 2020). The duration of the physical activity interventions lasted from 12 to 13 weeks in three studies (Davis 2012; Irandoust 2020; Nobre 2017), and the fourth study reported a 32-week intervention duration (Williams 2019).

All four studies implemented structured exercise training, which is understood as a specific and well-defined subset of physical activity [19]. For the purpose of this review, we categorised the interventions across the included studies as follows.

Aerobic exercise: we defined aerobic exercise as a dynamic physical activity in which the body's large muscles move rhythmically for a sustained period, increasing heart rate and breathing rate above resting levels to submaximal levels for a prolonged period. Examples of aerobic exercise include walking, running, swimming, and bicycling. We classified moderate-intensity continuous training (MICT) as a type of aerobic exercise.

MICT typically involves aerobic activities such as running, cycling, swimming, and rowing [71]. We defined the physical activity intervention offered in Williams 2019 as a type of MICT.

Combined training: Nobre 2017 implemented a progressive plyometric training programme across three levels of intensity (low-intensity lateral jump, moderate-intensity squat jump, and high-intensity jumps of increasing height), combined with plyometric speed and agility drills (e.g. accelerate, decelerate, and change direction). We categorised this programme as combined training, as any of the physical activity interventions involved a combination of two or more exercise modalities [72, 73]. Irandoust 2020 implemented a combination of video games, using devices such as Wii Fit, that included yoga, resistance training, aerobics, and balance mini-games. This study also utilised aerobic and resistance aquatic exercises. Similarly, Davis 2012 employed a combined training intervention that included running games and jump rope activities, along with resistance exercises derived from basketball and football movement patterns (e.g. squatting and core exercises).

We provide a detailed description of the interventions, including the frequency, intensity, type, time, volume, and progression (FITT-VP) parameters in Table 1. All studies reported adherence to the physical activity interventions. Two studies described the materials used in the physical activity interventions (Irandoust 2020; Williams 2019) (e.g. video game, equipment for running, ball games, and jump rope).

Comparators

Three of the four studies compared physical activity interventions with no intervention (Davis 2012; Irandoust 2020; Nobre 2017). Williams 2019 used sedentary activities as an 'attention control' comparison group.

Settings and providers

Three studies reported that physical activity interventions took place in schools (Irandoust 2020; Nobre 2017; Williams 2019); one study described the intervention as performed at the gym (Davis 2012). Three studies reported the supervision and providers of the interventions (e.g. researchers and instructors) (Irandoust 2020; Nobre 2017; Williams 2019).

Outcomes

The studies included in this review reported effect data for six critical outcomes and five important outcomes. The most frequently reported outcomes were BMI (Irandoust 2020; Nobre 2017; Williams 2019), and body fat percentage (Davis 2012; Nobre 2017; Williams 2019), followed by body weight and resistance to insulin reported in two studies (Davis 2012; Williams 2019).

Excluded studies

After evaluation of the full-text articles, we excluded 255 studies (44 studies were included in the 'sister' review in adolescents). The main reasons for exclusion were: ineligible intervention (88 studies) because the studies did not assess the effects of physical activity on its own; and ineligible population (85 studies) because the studies did not include children with obesity. For further details, see Supplementary material 3 and Figure 1.

Risk of bias in included studies

The risk of bias assessment was based on data from the primary published articles, supplemented by information received from study authors, the trial registry record, and the published protocol, where available. We assessed all four studies to be at an overall high risk of bias. We assessed two studies as having a low risk of bias in three domains (Irandoust 2020; Williams 2019), and two studies as having a low risk of bias in four domains (Davis 2012; Nobre 2017).

We assessed the included studies at the outcome level for blinding of participants and personnel, as well as blinding of outcome assessment.

For details on the risk of bias in the included studies, see Supplementary material 2. For an overview of the review authors' judgments about each risk of bias item for individual studies and across all studies, see Figure 2 and Figure 3.

Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies. The white bar indicates the domain was not applicable for three out of four included studies; only one study reported subjective outcomes.

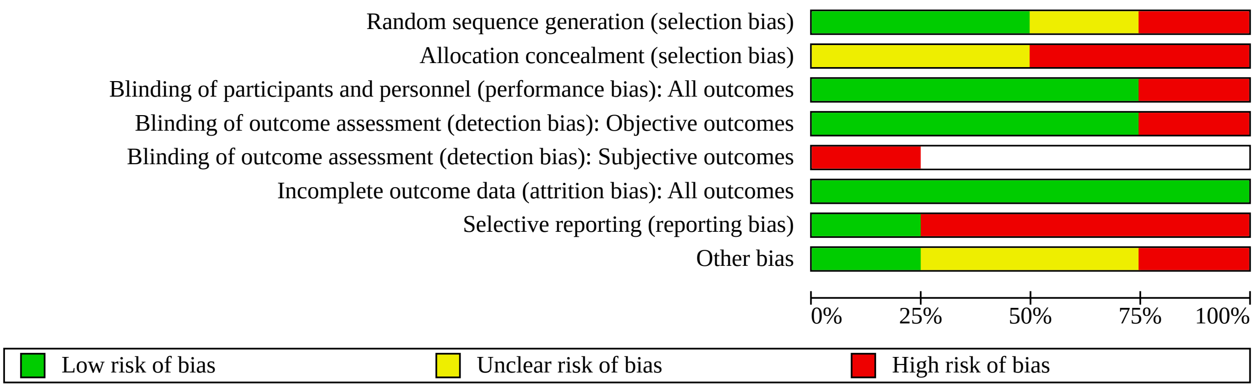


Figure 3. Risk of bias summary: review authors' judgements about each risk of bias item for each included study. The empty cells indicate that three out of four studies did not assess subjective outcomes.

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): Objective outcomes	Blinding of outcome assessment (detection bias): Subjective outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Davis 2012	+	−	+	+		+	−	?
Irandoust 2020	−	?	+	+		+	−	−
Nobre 2017	?	?	+	+		+	+	?
Williams 2019	+	−	−	−	−	+	−	+

Random sequence generation and allocation concealment (selection bias)

We judged two studies to have a low risk of selection bias for random sequence generation because they described adequate methods for this process (Davis 2012; Williams 2019). We judged one study to have an unclear risk of bias due to insufficient information (Nobre 2017). We rated only one study as high risk (Irandoust 2020).

With regard to allocation methods, we rated the risk of bias as unclear for two studies as they did not present sufficient information to allow definitive judgement (Irandoust 2020; Nobre 2017). The remaining studies did not use acceptable allocation methods, and we rated them as having a high risk in this domain (Davis 2012; Williams 2019).

Blinding

We divided the blinding domain into blinding of participants and personnel (performance bias) and blinding of outcome assessors (subjective and objective outcomes) (detection bias). As mentioned in Risk of bias assessment in included studies, in exercise studies, blinding of participants and care providers to treatment allocation is very rare.

Blinding of participants and personnel (performance bias)

Among the four studies included in this review, we rated blinding of participants and personnel as low risk in three studies (Davis 2012; Irandoust 2020; Nobre 2017), and high risk in one study (Williams 2019).

Blinding of outcome assessment (detection bias) – subjective outcomes

Three studies did not measure subjective outcomes and therefore this domain did not apply (Davis 2012; Irandoust 2020; Nobre 2017). We considered Williams 2019 to have a high risk of bias.

Blinding of outcome assessment (detection bias) – objective outcomes

With regard to blinding of assessor-reported outcomes, we judged three studies to be at low risk of detection bias, as they reported blinding of outcome assessors (Davis 2012; Irandoust 2020; Nobre 2017). We judged the risk of detection bias in Williams 2019 to be high because the assessor was not blinded.

Incomplete outcome data

We assessed all studies as having a low risk of attrition bias, as less than 20% of the randomised participants were lost to follow-up (Davis 2012; Irandoust 2020; Nobre 2017; Williams 2019)

Selective reporting

Among the four included studies, we found a study registry entry for two (Davis 2012; Williams 2019). After comparing the registry entry with the study report, we rated these studies as having a high risk of bias. We also judged Irandoust 2020 to have a high risk of bias in this domain. After comparing the methods with the results reported, we classified the remaining study as low risk of bias (Nobre 2017).

Other bias

We rated two studies as having an unclear risk because they provided insufficient information to assess whether an important risk of bias existed (Davis 2012; Nobre 2017). We did not find information on baseline inequities despite randomisation and missing data at baseline. We considered Irandoust 2020 to have a high risk of bias because the study protocol was not published, and we could not find missing details about ethics committee approval. We assessed the remaining study as having a low risk of bias (Williams 2019).

Synthesis of results

Our main comparison included four studies that compared physical activity of any form to control (Davis 2012; Irandoust 2020; Nobre 2017; Williams 2019), although the type of control varied. Three studies compared the intervention against no-exercise training interventions (Davis 2012; Irandoust 2020; Nobre 2017), while one study used sedentary activities for behaviour change (attention control) (Williams 2019).

Davis 2012 and Irandoust 2020 had three arms: two physical activity intervention groups and one control group. Irandoust 2020 permitted us to compare one form of combined training (aquatic exercises consisting of warm-up exercises, main exercises, and cool-down exercises) to another form of combined training (a variety of video games) (Irandoust 2020). Davis 2012 compared the effects of children doing 20 minutes versus 40 minutes of the same physical activities: running games, jumping rope, and resistance training exercises derived from basketball and soccer. Nobre 2017 compared a combined exercise programme involving progressively more advanced plyometric training (explosive jumping and power-based movements such as squat jumps), speed training, and agility drills (such as accelerating, decelerating, and changing direction) with a control group. We pooled all four studies in the meta-analysis for comparison 1.

See Summary of findings 1; Summary of findings 2; Summary of findings 3; Supplementary material 13; Supplementary material 14; Supplementary material 15; Supplementary material 9.

Comparison 1. Physical activity interventions versus control (non-exercise or behaviour change)

Critical outcomes

BMI (kg/m^2)

BMI (kg/m^2) at end of intervention

We pooled data from two studies in the meta-analysis of BMI (Irandoust 2020; Nobre 2017). Physical activity of any form may reduce BMI in children with obesity, but the evidence is very uncertain (MD −1.52 kg/m^2, 95% CI −2.74 to −0.29; I^2 = 0%; 2 studies, 118 children; very low-certainty evidence; Analysis 1.1).

Williams 2019 (175 children; very low-certainty evidence) found evidence of no difference between the physical activity arm (adjusted MD 0.4 kg/m^2, 95% CI 0.1 to 0.7) and the control arm (adjusted MD 0.8 kg/m^2, 95% CI 0.5 to 1.1).

BMI (kg/m^2) at follow-up

Only one study assessed BMI at 4 weeks of follow-up, implementing three study arms: two different types of combined training (video game-based and aquatic exercise-based) versus control (Irandoust 2020). These types of physical activity may reduce BMI at four weeks' follow-up in children with obesity, but the evidence is very uncertain (MD −2.45 kg/m^2, 95% CI −4.08 to −0.81; 1 study, 59 children; very low-certainty evidence; Analysis 1.2).

In addition, Williams 2019 found no evidence of a difference in BMI between groups at the 8- to 12-month follow-up (adjusted MD 0.27 kg/m^2, 95% CI −0.3 to 0.8; P = 0.34; 175 children; very low-certainty evidence).

BMI z-score

BMI z-score at end of intervention

Two studies assessed BMI z-scores (Davis 2012; Williams 2019). Differences in effect estimates and adjusted analyses precluded a pooled statistical synthesis of the results.

Davis 2012 had three arms and two were active (20-minute and 40-minute aerobic exercise programmes involving the same combination of activities). Very low-certainty evidence suggests that low- or high-dose combined training has little to no effect on BMI z-scores in children with obesity (MD −0.10 z-score units, 95% CI −0.22 to 0.02; I^2 = 29%; 222 children; Analysis 1.3).

Williams 2019 (175 children; very low-certainty evidence) reported within-group changes from baseline to post-test (i.e. immediately after the intervention ended), adjusted for cohort, race, and sex in mixed repeated-measures analysis of variance (ANOVA). The study provided evidence of no difference between the physical activity (adjusted MD 0.4 SD, 95% CI 0.1 to 0.7) and the control arm (adjusted MD 0.8 SD, 95% CI 0.5 to 1.1).

BMI z-score at 8- to 12-month follow-up

Williams 2019 found no evidence of a difference in BMI z-scores between groups at the 8- to 12-month follow-up (adjusted MD 0.02 SD, 95% CI −0.05 to 0.09; P = 0.54; 175 children; very low-certainty evidence).

Body weight (kg)

Body weight at end of intervention

Two studies provided data for body weight (Irandoust 2020; Nobre 2017). Very low-certainty evidence means that it is uncertain whether physical activity (video game-based, aquatic exercise-based, and plyometric-based) reduces body weight compared to control in children with obesity (MD −0.86 kg, 95% CI −3.17 to 1.46; I^2 = 0%; 118 children; Analysis 1.4).

Body weight (kg) at 4-week follow-up

One study assessed body weight (kg) at four weeks' follow-up (Irandoust 2020). Very low-certainty evidence means that it is uncertain whether physical activity (video games and aquatic exercise) reduces body weight at four weeks' follow-up compared to control in children with obesity (MD −1.03 kg, 95% CI −3.74 to 1.69; 59 children; Analysis 1.5).

Health-related quality of life at end of intervention

One study assessed health-related quality of life (Williams 2019). There is very low-certainty evidence about whether aerobic exercise, consisting of moderate-intensity continuous training (MICT), has an effect on health-related quality of life compared to control in children with obesity (MD −0.60 points, 95% CI −4.22 to 3.02; 175 children; Analysis 1.11)

Adiposity and fat distribution

Body fat percentage at end of intervention

Three of the four studies provided data on body fat percentage (Davis 2012; Nobre 2017; Williams 2019). Physical activity compared with control may result in little to no difference in body fat percentage in children with obesity, but the evidence is very uncertain (MD −1.23%, 95% CI −2.50 to 0.03; I^2 = 0%; 456 children; very low-certainty evidence; Analysis 1.12).

Body fat percentage at 8- to 12-month follow-up

Williams 2019 found no evidence of a difference in body fat percentage between groups at the 8- to 12-month follow-up (adjusted MD 0.37%, 95% CI −0.05 to 0.09; P = 0.54; 175 children; very low-certainty evidence).

Sum of skinfolds at end of intervention

One study assessed the sum of skinfolds (Nobre 2017). Very low-certainty evidence means that it is uncertain whether physical activity reduces the sum of skinfolds compared to control in children with obesity (MD −1.80 mm, 95% CI −8.88 to 5.28; 59 children; Analysis 1.13).

Visceral fat at end of intervention

Davis 2012 (222 children; very low-certainty evidence) expressed adiposity distribution as visceral fat, which was measured using magnetic resonance images. The study found larger reductions between baseline and post-test for the high-dose physical activity arm than the control arm (visceral fat, adjusted MD −3.86 cm^3, 95% CI −5.98 to −1.73; P < 0.001). Similar reductions were observed in the low-dose physical activity arm with respect to the control arm (visceral fat, adjusted MD −2.77 cm^3, 95% CI −4.91 to −0.63; P = 0.01).

Waist circumference at end of intervention

Williams 2019 (175 children; very low-certainty evidence) measured waist circumference between the lowest rib and the iliac crest. The study reported within-group changes from baseline to post-test adjusted for cohort, race, and sex in mixed repeated-measures ANOVA. The study provided evidence of no difference between the physical activity (adjusted MD 2.1 cm, 95% CI 1.1 to 3.2) and the control arm (adjusted MD 2 cm, 95% CI 1 to 3.1).

Waist circumference at 8- to 12-month follow-up

Williams 2019 found no evidence of a between-group difference in waist circumference at the 8- to 12-month follow-up (adjusted MD 1.6 cm, 95% CI −0.1 to 3.3; P = 0.07; 175 children; very low-certainty evidence).

Glycaemia

None of the included studies reported this outcome.

Adverse events

Minor adverse events at end of intervention

Davis 2012 provided data for adverse events. Physical activity (20-minute and 40-minute aerobic exercise programmes involving the same combination of activities with low- or high-dosage intervention (i.e. different total volumes in minutes per session)) compared to control may increase the number of minor adverse events, but the evidence is very uncertain (risk ratio (RR) 3.58, 95% CI 1.95 to 6.55; number needed to treat for an additional harmful outcome (NNTH): 3, 95% CI 1 to 8; 222 children; very low-certainty evidence; Analysis 1.14).

Serious adverse events at end of intervention

The evidence is very uncertain about the effect of physical activity on serious adverse events (RR 1.08, 95% CI 0.10 to 11.76; 222 children; very low-certainty evidence; Analysis 1.15) in children with obesity when compared to control.

Important outcomes

Physical well-being

None of the included studies reported this outcome.

Mental well-being

Williams 2019 was the only study that provided data on mental well-being, reporting data on depression (assessed with the Children's Depression Inventory instrument), anger expression (assessed with the Pediatric Anger Expression Scale), and global self-worth (assessed with the Harter Self-Perception Profile for Children).

Depression at end of intervention

Williams 2019 used the Children's Depression Inventory to assess depressive symptoms (continuous outcome), to classify whether children 'screened positive' for clinically concerning symptoms (a derived dichotomous outcome based on a cut-off score), and risk of self-harm (dichotomous outcome).

- It is uncertain whether physical activity (aerobic exercise–MICT) has an effect on depressive symptoms compared to control in children with obesity (MD −0.50, 95% CI −2.15 to 1.15; 175 children; very low-certainty evidence; Analysis 1.6). It is uncertain whether physical activity reduced depressive symptoms in females (MD −0.85, 95% CI −1.06 to −0.64; 107 children; Analysis 1.6) and males (MD 0.90, 95% CI 0.59 to 1.21; 68 children; Analysis 1.6) because the certainty of evidence is very low.
- It is uncertain whether physical activity (aerobic exercise–MICT) reduces the number of children who screen positive for clinically concerning symptoms compared to control because the certainty of evidence is very low (RR 0.71, 95% CI 0.16 to 3.07; 175 children; Analysis 1.9).
- Physical activity (aerobic exercise–MICT) may have little to no effect on reducing the risk of self-harm, but the evidence is very uncertain (RR 0.65, 95% CI 0.29 to 1.45; 175 children; Analysis 1.10).

Anger expression at end of intervention

Williams 2019 applied the Pediatric Anger Expression Scale and subscales, including 'anger-in', which assesses suppression or internalisation of anger, 'anger-out', outward expression of anger, and 'anger control', which assesses ability to regulate anger.

- Compared to control, physical activity (aerobic exercise–MICT) may result in little to no difference in anger expression in children with obesity, but the evidence is very uncertain (MD −0.30, 95% CI −1.31 to 0.71; 175 children; Analysis 1.7).
- It is uncertain whether physical activity (aerobic exercise–MICT) reduces anger-in expression compared to control because the certainty of evidence is very low (MD 0.00, 95% CI −0.76 to 0.76; 175 children; Analysis 1.7).
- Compared to control, physical activity (aerobic exercise–MICT) may slightly reduce anger-out expression, but the evidence is very uncertain (MD −1.00, 95% CI −1.59 to −0.41; 175 children; Analysis 1.7).
- Physical activity (aerobic exercise–MICT, low or high dose) may result in little to no difference in anger control, but the evidence is very uncertain (MD 0.00, 95% CI −0.59 to 0.59; 175 children; Analysis 1.7).

Global self-worth at end of intervention

Compared with control, physical activity (aerobic exercise–MICT) may result in little to no difference in global sense of self-worth, as measured with the Harter Self-perception Profile for Children, in children with obesity, but the evidence is very uncertain (MD 0.10, 95% CI −0.07 to 0.27; 175 children; Analysis 1.8).

Physical activity levels at end of intervention

Based on evidence from Williams 2019 (175 children), it is very uncertain whether physical activity (aerobic exercise–MICT) has an effect on moderate-to-vigorous physical activity levels, measured as minutes per day (min/day), compared to control in children with obesity. This study reported within-group changes from baseline to post-test (i.e. immediately after the intervention ended) adjusted for cohort, race, and sex in mixed repeated-measures ANOVA. We could not retrieve raw data from the authors. The physical activity arm showed an increase of 2.1 min/day (95% CI −2.3 to 6.5), whereas the control arm showed a larger increase (MD 4.0 min/day, 95% CI −0.4 to 8.4).

Blood pressure at end of intervention

Based on evidence from Williams 2019 (175 children), it is very uncertain whether physical activity (aerobic exercise–MICT) reduces blood pressure (systolic and diastolic pressure) when compared to control in children with obesity. The authors reported within-group changes from baseline to post-test adjusted for cohort, race, and sex in mixed repeated-measures ANOVA without differences between baseline and post-test. However, we could not retrieve raw data from the authors (see Table 2).

Hyperinsulinemia

None of the included studies reported this outcome.

Resistance to insulin at end of intervention

Two studies (397 children) provided data for resistance to insulin (Davis 2012; Williams 2019). Very low-certainty evidence means that it is uncertain whether physical activity has an effect on resistance to insulin in children with obesity. The difference in effect estimates and adjusted analyses precluded a pooled statistical synthesis.

Davis 2012 (222 children; very low-certainty evidence) expressed resistance to insulin as the insulin area under the curve (AUC), which was calculated using the trapezoidal rule. The study found larger reductions between baseline and post-test for the high-dose physical activity arm than the control arm (AUC, adjusted MD −3.56 μU/mL×103, 95% CI −6.26 to −0.85; P = 0.01). Similar reductions were observed in the low-dose physical activity arm with respect to the control arm (AUC, adjusted MD −2.96 µU/mL×103, 95% CI −5.69 to −0.22; P = 0.03).

Williams 2019 (175 children; very low-certainty evidence) measured resistance to insulin using the Homeostatic Model Assessment of Insulin Resistance (HOMA-IR) index. The study reported within-group changes from baseline to post-test adjusted for cohort, race, and sex in mixed repeated-measures ANOVA. The study provided evidence of no difference between the physical activity (MD 0.1 HOMA units, 95% CI −0.3 to 0.4) and the control arm (MD −0.0 HOMA units, 95% CI −0.4 to 0.3). We could not retrieve raw data from the authors to investigate further.

Alterations in lipid metabolism

One study (175 children; very low-certainty evidence) provided data for alterations in lipid metabolism (Williams 2019). Williams 2019 reported within-group changes from baseline to post-test adjusted for cohort, race, and sex in mixed repeated-measures ANOVA. However, we could not retrieve raw data from the authors to investigate further. The study effect estimates are shown in Table 3.

Total cholesterol at end of intervention

Very low-certainty evidence means that it is uncertain whether physical activity has an effect on total cholesterol when compared to control in children with obesity (Williams 2019).

High-density lipoprotein (HDL) at end of intervention

Physical activity increased HDL significantly compared to the control group. However, because of the very low certainty of the evidence, we had very little confidence in this effect estimate (Williams 2019).

Low-density lipoprotein (LDL) at end of intervention

Very low-certainty evidence means that it is uncertain whether physical activity has an effect on LDL when compared to control in children with obesity (Williams 2019).

Triglycerides at end of intervention

It is uncertain whether physical activity reduced triglycerides compared to control in children with obesity because the certainty of evidence is very low (Williams 2019).

Remaining important outcomes

None of the included studies reported the following important outcomes: presence of obesity comorbidities; obesity-associated disability; lipid hormones; alterations in hunger and satiety; disability in any of the functionality domains; mortality; prevalence of obesity in adulthood; and access to health services.

Comparison 2. Combined training (aquatic exercises) versus combined training (video game exercises)

Critical outcomes

Irandoust 2020 provided data only for BMI and body weight at the end of intervention and at four weeks of follow-up. See Summary of findings 2; Supplementary material 14. Irandoust 2020 did not assess any of the other critical and important outcomes.

BMI (kg/m^2)

BMI (kg/m^2) at end of intervention

It is uncertain whether combined training in an aquatic setting reduced BMI compared to combined training performed with video games in children with obesity because the certainty of evidence is very low (MD −0.90 kg/m^2, 95% CI −3.21 to 1.41; 39 children; Analysis 2.1).

BMI (kg/m^2) at 4-week follow-up

Very low-certainty evidence showed that it is uncertain whether combined training in an aquatic setting compared to combined training performed with video games in children with obesity reduces BMI at four weeks' follow-up (MD 0.03 kg/m^2, 95% CI −2.54 to 2.60; 39 children; Analysis 2.2)

Body weight

Body weight (kg) at end of intervention

It is uncertain whether combined training in an aquatic setting reduced body weight compared to combined training performed with video games in children with obesity because the certainty of evidence is very low (MD −1.30 kg/m^2, 95% CI −5.66 to 3.06; 39 children; Analysis 2.3).

Body weight (kg) at 4-week follow-up

Very low-certainty evidence indicated that combined training in an aquatic setting compared to combined training performed with video games may result in little to no difference in body weight in children with obesity at four weeks' follow-up (MD −0.70%, 95% CI −5.70 to 4.30; 39 children; Analysis 2.4).

Comparison 3. Low-dose combined training versus high-dose combined training

Critical outcomes

Davis 2012 provided data for BMI z-score, body fat percentage, fasting glucose, and adverse events (minor and serious). See Summary of findings 3; Supplementary material 15. Davis 2012 did not assess any of the other critical and important outcomes.

BMI z-score

BMI z-score at end of intervention

High-dose combined training may slightly reduce BMI z-score in children with obesity compared to low-dose combined training, but the evidence is very uncertain (MD 0.12 z-score units, 95% CI 0.10 to 0.14; 144 children; Analysis 3.1).

Adiposity and fat distribution

Body fat percentage at end of intervention

It is uncertain whether high-dose combined training compared to low-dose combined training reduces body fat percentage because the certainty of the evidence was very low (MD 1.07%, 95% CI −0.74 to 2.88; 144 children; Analysis 3.2).

Glycaemia

Fasting glucose at end of intervention

High-dose combined training compared to low-dose combined training may result in little to no difference in fasting glucose in children with obesity, but the evidence is very uncertain (MD −0.50%, 95% CI −2.77 to 1.77; 144 children; Analysis 3.3).

Adverse events

Minor adverse events at end of intervention

Very low-certainty evidence showed that it is uncertain whether low-dose combined training reduces the number of minor adverse events (RR 0.91, 95% CI 0.64 to 1.30; 144 children; Analysis 3.4) when compared to high-dose combined training in children with obesity. To access the analysis, please see Supplementary material 5.

Serious adverse events at end of intervention

The evidence is very uncertain about the effect of low-dose combined training on serious adverse events (RR 3.08, 95% CI 0.13 to 74.45; 144 children; very low-certainty evidence; Analysis 3.5) in children with obesity when compared to high-dose combined training (Supplementary material 5).

Equity assessment

This review sought to identify studies that had reported on characteristics known to be important from an equity and disadvantage perspective, using the PROGRESS (Place, Race, Occupation, Gender, Religion, Education, Socioeconomic status (SES), Social status) framework [61]. Where reported, interventions did not appear to increase health inequalities. We recorded where interventions and outcomes were analysed by any of the eight PROGRESS categories. For gender, one study reported outcomes analysed by gender (Williams 2019). Two studies assessed outcomes in Black children (Davis 2012; Williams 2019), and one in Hispanic children (Davis 2012), and all studies investigated the outcome in the 8- to 9-year-old age group (Davis 2012; Irandoust 2020; Nobre 2017; Williams 2019). PROGRESS characteristics are reported in Supplementary material 8.

Reporting biases

A librarian developed and conducted a comprehensive systematic search of the literature. In addition, a pair of review authors searched grey literature resources (e.g. reports, dissertations, theses, and conference abstracts), and we did not prespecify any language restrictions. We detected no industry influence on the included studies.

We could not conduct our planned formal assessment of publication bias (e.g. funnel plot, Egger's test, or both) because there were too few studies. We assessed potential selective outcome reporting within studies by comparing the number and order of outcomes reported in trial registry entries, study protocols, or methods sections with those reported in the final published articles. We identified discrepancies between the prespecified and published outcomes in three studies (Davis 2012; Irandoust 2020; Williams 2019).

DISCUSSION

This systematic review included four RCTs investigating different physical activity interventions aimed at improving obesity-related health outcomes in children with obesity. The children's mean age was 9 years, and all children were living with obesity (BMI > 95th percentile) at baseline. Only sparse information was available on comorbidities, and none of the studies included children with disabilities. Two studies were conducted in the USA, one in Brazil, and one in Iran. Three studies (75%) compared physical activity interventions with no-exercise training interventions (Davis 2012; Irandoust 2020; Nobre 2017), and one study compared physical activity with behaviour change (Williams 2019). Generally, the physical activity interventions had at least one aerobic exercise arm (Davis 2012; Irandoust 2020; Williams 2019), lasted 12 to 13 weeks (Davis 2012; Irandoust 2020; Nobre 2017), took place in schools (Irandoust 2020; Nobre 2017; Williams 2019), and were supervised by researchers and instructors (Irandoust 2020; Nobre 2017; Williams 2019). None of the studies provided details about the theoretical approach behind the physical activity interventions. The most frequently reported outcomes were BMI, body weight, body fat percentage, and resistance to insulin.

Summary of main results

We are uncertain of the effect of physical activity interventions (any form) due to very low-certainty evidence, which suggests little to no difference in most of the obesity-related health outcomes of children up to the age of 9 years with obesity when compared to control.

Physical activity interventions versus control (non-exercise or behaviour change)

The evidence is very uncertain about the effect of physical activity (any form) compared to control. Very low-certainty evidence suggests that physical activity of any form may reduce BMI in children with obesity (2 studies, 118 children). For BMI measured as changes in z-score, the evidence is very uncertain about the effect of physical activity (any form) on the BMI z-score of children up to the age of 9 years with obesity (2 studies, 397 children; very low-certainty evidence) compared to control.

For weight reduction, very low-certainty evidence suggests that physical activity of any form may result in little to no difference in body weight in children with obesity aged up to 9 years (2 studies, 118 children).

The evidence is very uncertain about the effect of physical activity (any form) on the health-related quality of life of children up to the age of 9 years with obesity (1 study, 175 children; very low-certainty evidence) compared to control.

For adiposity and fat distribution, very low-certainty evidence suggests that physical activity of any form may result in little to no difference in body fat percentage in children with obesity aged up to 9 years (3 studies, 456 children).

The evidence regarding adverse events is very uncertain. Compared with control groups, physical activity may increase the risk of minor adverse events, although this finding is based on very low-certainty evidence from a single study involving 222 children. Similarly, very low-certainty evidence from the same study suggests that physical activity interventions may result in little to no difference in serious adverse events among children with obesity aged up to 9 years.

Combined training versus combined training

We obtained very low-certainty evidence for other active comparisons, including combined training (aquatic exercises versus video game exercises) (Irandoust 2020), and dose variation training (low-dose versus high-dose) (Davis 2012).

Very low-certainty evidence suggests that combined training (aquatic exercises or video game exercises) may result in little to no difference in BMI or body weight, with both outcomes assessed at the end of the intervention and at 4-week follow-up in children with obesity aged up to 9 years (1 study, 39 children).

The evidence is very uncertain, but high-dose combined training may reduce BMI z-score more effectively than low-dose combined training in children with obesity up to 9 years of age (1 study, 144 children; very low-certainty evidence). Very low-certainty evidence suggests that combined training (low-dose or high-dose) may result in little to no difference in body fat percentage, fasting glucose, and minor or serious adverse events in children up to the age of 9 years with obesity (1 study, 144 children).

Limitations of the evidence included in the review

We used GRADE to assess the certainty of the evidence. We downgraded the certainty of this body of evidence owing to several methodological limitations, a small sample size, and wide confidence intervals. Reasons for downgrading for each of the GRADE criteria are reported below. We could not explore subgroup effects given the small number of studies available.

Risk of bias

Overall, the risk of bias amongst the included studies was high or unclear for selection, performance, detection, attrition, selective reporting, and other sources of bias. We assessed studies with baseline imbalances and with missing details about ethics committee approval as having an unclear or high risk for other biases. We judged three studies to be at high risk for selective reporting bias (Davis 2012; Irandoust 2020; Nobre 2017).

Imprecision

We downgraded the certainty of the evidence for most of the outcomes investigated in this review. Overall, the number of children included in each meta-analysis ranged from moderate (100 to 300) to small (< 100). All the outcomes were informed by four or fewer studies.

Inconsistency

We did not downgrade the certainty of the evidence for any of the outcomes due to inconsistency (e.g. the point estimates did not vary widely, and the confidence intervals substantially overlapped).

Indirectness

We did not downgrade the certainty of the evidence for any of the outcomes due to indirectness (e.g. the population and the intervention were highly applicable for the decision context).

Publication bias

We did not downgrade the certainty of the evidence for any of the outcomes due to publication bias (e.g. a comprehensive search was conducted, grey literature searched, and no restrictions applied to study selection based on language).

Limitations of the review processes

We revised our original eligibility criteria to include studies with a minimum intervention duration of 12 weeks, regardless of the post-intervention follow-up period. We listed three studies published in Persian and two studies published in Chinese as 'awaiting classification' due to time constraints. External colleagues are helping us determine whether these studies are eligible; if they are, we will integrate evidence from these studies in a review update. We appraised the included studies with Cochrane's RoB 1 tool [48]. Although unlikely, we cannot rule out that different judgements might have been reached if we had applied the most recent tool, RoB 2 [74]. We restricted eligibility to studies that included children with obesity at baseline, which may have increased the risk of selection and selective non-reporting bias, if some studies provided insufficient information on baseline obesity status (i.e. eligible studies may have been missed). However, the broad range of outcomes considered strengthened the review's ability to address the research question and provide a comprehensive evidence base. This review was strengthened by the use of predefined research methods. We searched a range of bibliographic databases without language restrictions, clinical trial registries to identify unpublished trial data, and Google Scholar to identify grey literature. These strategies were intended to minimise selective outcome reporting and improve the likelihood of identifying additional data not available in the published journal literature. We deployed standardised procedures for study selection, and trained review authors in data extraction and risk of bias assessment through a standardised process. In addition, the review team had expertise in library science, evidence synthesis for evidence-informed decision-making, exercise prescription for non-communicable conditions, physiotherapy, and knowledge translation.

Agreements and disagreements with other studies or reviews

Comparing the results of this systematic review with those of earlier reviews is challenging because most have focused on primary prevention of obesity (i.e. children without obesity). In addition, some reviews have addressed multimodal interventions, such as diet plus physical activity [75, 76]. A 2017 Cochrane review by Mead and colleagues found low-certainty or very low-certainty evidence supporting behaviour-changing interventions (behavioural, diet, and physical activity components) for the treatment of obesity in children (aged 6 to 11 years) [45]. In agreement with our GRADE assessments, Mead and colleagues downgraded their certainty in the evidence due to methodological limitations, inconsistency, and imprecision in the included studies. Similarly to our findings, low-certainty evidence from the WHO guidelines on physical activity showed no consistent effect of high-intensity interval training or resistance training on BMI, body fat percentage, waist circumference, or fat mass among children and adolescents without obesity [29]. Brown and colleagues found that childcare- and school-based physical activity interventions may not reduce BMI or BMI z-score in children aged 0 to 5 years with overweight and without obesity [7].

AUTHORS' CONCLUSIONS

Implications for practice

Drawing implications for practice from the randomised controlled trials (RCTs) included in this review is not straightforward, given the heterogeneity present across the physical activity interventions, especially in terms of the training modes and implementation procedures, as well as the scarcity of information available for some outcomes, such as health-related quality of life, mental well-being, and adverse events. This review yielded very low-certainty evidence suggesting that any form of physical activity for around 12 weeks, mainly in school-based settings, results in little to no difference in health outcomes of children aged 9 years with obesity when compared to no-exercise training interventions or a behavioural change.

Overall, the findings from this review are limited to children up to the age of 9 years with obesity who engage in school-based physical activity interventions for around 12 weeks.

Equity-related implications for practice

Despite the lack of information on sociodemographic data (e.g. ethnicity and socioeconomic status) and comorbidities, we observed differences in contextual factors across the RCTs. For example, the RCTs came from three very different countries (i.e. the USA (Davis 2012; Williams 2019), Brazil (Nobre 2017), and Iran (Irandoust 2020)); two studies assessed outcomes in Black children (Davis 2012; Williams 2019), and one in Hispanic children (Davis 2012). The limited amount of information available from the RCTs precluded us from exploring subgroup analyses by age, gender, place of residence, socioeconomic status, or other characteristics.

Implications for research

Higher-quality and better-reported RCTs are needed as the current evidence is of low methodological quality. Such future research would likely change our findings. We categorised 29 studies as 'awaiting classification' (including seven conference abstracts and 12 trial registry records); incorporating the results of these studies when they become available may strengthen the evidence base. Besides ensuring adequate randomisation procedures, future RCTs should provide clear details on the diagnostic criteria used for obesity and should adhere to reporting checklists, such as CONSORT [77], TIDieR [78], and CERT [79]. For instance, none of the included studies assessed intervention fidelity (TIDieR item 11; CONSORT 4C; CERT item 16a) to ensure the intervention delivery matched the protocol (Davis 2012; Irandoust 2020; Nobre 2017; Williams 2019). Only Irandoust 2020 and Williams 2019 reported the specific material used in the physical activity interventions (TIDieR item 3 and CERT item 8), which is necessary for accurate replication. In addition, most studies provide insufficient detail on the components and structure of the control groups (TIDieR items 4 and 8, CONSORT item 4) (Davis 2012; Irandoust 2020; Nobre 2017). Furthermore, long-term effects and safety data for physical activity interventions in children with obesity should also be collected.

Very low-certainty evidence from four RCTs showed that it is uncertain whether physical activity has an effect on obesity-related health outcomes in children with obesity. This evidence base is constrained by serious methodological limitations, clinical heterogeneity, as well as small-study effects and imprecise results. Important knowledge gaps remain as none of the included RCTs enroled children with disabilities, the RCTs provided little information on contextual factors, and the amount of evidence hindered the analysis of relevant subgroups, such as physical activity modes or intervention length. Given the low methodological quality of current studies, more rigorous and transparently reported RCTs are essential. Such research is highly likely to have an important impact on our confidence in the estimates of effect and will likely change our findings. Future RCTs should also consider investigating important obesity-related health outcomes, including physical well-being, presence of obesity co-existing complications and disabilities, hyperinsulinemia, lipid hormones, and access to health services.

Equity-related implications for research

None of the studies included in this review specifically examined how equity-related variables moderate the effects of physical activity interventions on children with obesity. Future randomised trials should address this gap by exploring how such factors may mediate or moderate intervention outcomes. The PROGRESS framework offers a useful starting point, including key variables such as place of residence, race, ethnicity, culture, language, gender/sex, religion, education, and socioeconomic status [61].

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: 10.1002/14651858.CD015988.

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Analyses

Supplementary material 6 Data package

Supplementary material 7 Record of study investigators providing information on included studies

Supplementary material 8 Description of PROGRESS characteristics

Supplementary material 9 Evidence profile: physical activity (any form) versus control (non-exercise or behavior change)#

Supplementary material 10 Checklist to aid consistency and reproducibility of GRADE assessments: physical activity (any form) versus control (non-exercise or behavior change)

Supplementary material 11 Checklist to aid consistency and reproducibility of GRADE assessments: Combined training (aquatic exercises) versus combined training (video games exercises)

Supplementary material 12 Checklist to aid consistency and reproducibility of GRADE assessments: Combined training (low dose) versus combined training (high dose)

Supplementary material 13 Evidence profile: physical activity (any form) versus control (non-exercise or behaviour change)

Supplementary material 14 Evidence profile: Combined training (aquatic exercises) versus combined training (video games exercises)

Supplementary material 15 Evidence profile: Combined training (low dose) versus combined training (high dose)

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

Cochrane Metabolic and Endocrine Disorders supported the authors in the development of this review. The following people conducted the editorial process for this review:

- Sign-off Editor (final editorial decision): Toby Lasserson, Cochrane;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Joey Kwong, Cochrane Central Editorial Service and Jenny Bellorini, Cochrane Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments and supported the editorial team): Lisa Wydrzynski, Cochrane Central Editorial Service;
- Copy Editor (copy editing and production): Faith Armitage, Cochrane Central Production Service;
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Contributions of authors

- Andrés F. Loaiza-Betancur: conceptualisation; data curation; investigation (study selection); methodology; project administration; resources; software; supervision; validation; visualisation; writing—original draft; writing—review and editing.
- Lisette Ethel Iglesias-Gonzalez: data curation; investigation (study selection); validation; visualisation; writing—original draft; writing—review and editing.
- Nathaly Chavez Guapo: data curation; investigation (study selection); validation; visualisation; writing—original draft; writing—review and editing.
- Camila Escobar Liquitay: literature searches; visualisation; writing—review and editing.
- Julia Bidonde: overall methodological guidance; writing—review and editing.

Declarations of interest

JB: none known.

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CMEL: none known.

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AFLB: no relevant interests; WHO (grant/contract) – this review was conducted to inform one or more guideline recommendations on the 'Integrated management of adolescents 10 to 19 years of age with obesity in all their diversity: a primary health care approach for improved health, functioning and reduced obesity-associated disability'. The Department of Nutrition and Food Safety at the WHO commissioned and provided financial support for this work.

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Registration and protocol

The full protocol was registered, and it is publicly available (https://osf.io/anqb8).

Data, code and other materials

The data package from this systematic review is available in Supplementary material 6. All data extracted from the included studies are also available from the corresponding author upon request.

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ADDITIONAL TABLES

Table 1. Overview of syntheses and included studies

Study ID (Country)	Participants	Interventions and control	Outcomes reported
Davis 2012 (USA) See also NCT00108901	n = 222 (128 female) Exp 1: n = 71, age (yr) = 9.3 (SD 0.9) Exp 2: n = 73, age (yr) = 9.4 (SD 1.2) Con: n = 78, age (yr) = 9.4 (SD 1.1)	**Physical activity:** combined training – low dose (running games, jump rope, and resistance training exercise derived from basketball and soccer) FITT-VP: Frequency: 5/week; Intensity: vigorous activity; Time: 13 weeks; Volume: 20 min per session; Progression: NR **Physical activity:** combined training – high dose (running games, jump rope, and resistance training exercise derived from basketball and soccer) FITT-VP: Frequency: 5/week; Intensity: vigorous activity; Time: 13 weeks; Volume: 40 min per session. Progression: NR **Control:** no exercise training intervention	BMI z-score, body fat percentage, visceral fat, fasting glucose, adverse events (minor and serious), and resistance to insulin
Irandoust 2020 (Islamic Republic of Iran)	n = 61 (0 female) Exp 1: n = 21, age (yr) = 8.91 (SD 1.21)	**Physical activity:** combined training (yoga, resistance training, aerobics, and balance video games) FITT-VP: Frequency: 3/week, Intensity: 50–70% HRR (11-13 RPE), Time: 12 weeks; Volume: 60 min per	BMI and body weight

Table 1. Overview of syntheses and included studies *(Continued)*

	Exp 2: n = 20, age (yr) = 9.30 (SD 1.30) Con: n = 20 age (yr) = 8.95 (SD 1.15)	session; Progression: NR **Physical activity:** combined training (aerobic and resistance exercise; aquatic exercise) FITT-VP: Frequency: 3/week, Intensity: 40–70% HRR (11-13 RPE), Time: 12 weeks; Volume: 60 min per session; Progression: NR **Control:** no exercise training intervention	
Nobre 2017 (Brazil)	n = 59 (0 female) Exp: n = 40, age (yr) = 9.8 (SE 0.9) Con: n = 19, age (yr) = 9.9 (SE 1.1)	**Physical activity:** combined training (plyometric training levels 1, 2, and 3 combined with agility drills) FITT-VP: Frequency: 2/week; Intensity: low, moderate to high; Time: 12 weeks; Volume: 10 to 24 sets, per 50 to 120 jumps, full recovery between sets; Progression: volume increased every 2 weeks **Control:** no exercise training intervention	BMI, body weight, body fat percentage, and sum of skinfolds
Williams 2019 (USA) See also NCT02227095, NCT02383485	n = 175 (107 female) Exp: n = 90, age (yr) = 9.7 (SD 0.88) Con: n = 85, age (yr) = 9.6 (SD 0.88)	**Physical activity:** AE-MICT FITT-VP: Frequency: 5/week; Intensity: vigorous activity; Time: 32 weeks; Volume: 40 min per session; Progression: NR **Control:** behaviour change	BMI, BMI z-score, health-related quality of life, body fat percentage, waist circumference, mental well-being (depressive symptoms, positive screen, self-harm, depression, anger (expression, in-, out-, and control)), and self-perception – self-worth

Abbreviations: AE: aerobic exercise; BMI: body mass index; Con: control; Exp: experimental; FITT-VP: Frequency, Intensity, Type, Time, Volume, Progression; HRR: reserve heart rate; MICT: moderate-intensity continuous training; min: minutes; NR: not reported; reps: repetition; RPE: rate of perceived exertion; SD: standard deviation; SE: standard error; yr: years

Table 2. Physical activity (any mode) versus control: blood pressure

Blood pressure (mmHg)	Physical activity arm[a]	Control arm[a]
Systolic blood pressure	MD 0.1 mmHg (95% CI −1.5 to 1.7)	MD 0.3 mmHg (95% CI −1.3 to 2.0)
Diastolic blood pressure	MD 1.1 mmHg (95% CI −0.1 to 2.2)	MD 0.0 mmHg (95% CI −1.1 to 1.2)

CI: confidence interval; **MD:** mean difference, **mmHG:** millimetres of mercury
[a]Effect estimates correspond to within-group changes from baseline to post-test adjusted for cohort, race, and sex in mixed repeated-measures analysis of variance (ANOVA).
*Statistically significant within-group differences (P < 0.05).
**Statistically significant between-group differences (P < 0.05).

Table 3. Physical activity (any mode) versus control: blood lipid metabolism

Blood lipids (mg/dL)	Physical activity arm[a]	Control arm[a]

Table 3. Physical activity (any mode) versus control: blood lipid metabolism *(Continued)*

Total cholesterol	MD −6.0 (95% CI −10.4 to −1.7)*	MD −5.1 (95% CI −9.2 to −1.0)*
HDL**	MD 5.0 (95% CI 2.9 to 7.2)*	MD −1.1 (95% CI −3.2 to 0.9)
LDL	MD −6.0 (95% CI −10.4 to −1.7)*	MD −5.1 (95% CI −9.2 to −1.0)*
Triglycerides	MD −7.0 (95% CI −17 to 3.2)	MD −9.1 (95% CI −1.9 to 0.7)

CI: confidence interval; **HDL:** high-density lipoprotein; **LDL:** low-density lipoprotein; **MD:** mean difference
[a]Effect estimates correspond to within-group changes from baseline to post-test adjusted for cohort, race, and sex in mixed repeated-measures analysis of variance (ANOVA).
*Statistically significant within-group differences (P < 0.05).
**Statistically significant between-group differences (P < 0.05).